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Objectives

- Mobile technologies.
- Android overview.
- Open Handset Alliance.
- Android Architecture (diagram).
- Why Android?

Introduction to Mobile Technologies

Topics

- Evolution of Mobile Technology.
- Mobile OS overview (Android, iOS).
- Need of Mobile Applications.

Mobile Technology

Mobile Technology refers to the technology that enables wireless communication and computing using portable devices such as smartphones, tablets, and wearable gadgets. It allows users to communicate, access the internet, run applications, and exchange data anytime and anywhere without being tied to a fixed location.

Evolution of Mobile Technology

1G (First Generation)

- Only voice calling.
- Analog technology.
- No data service.

2G

- Digital communication.
- SMS, MMS start.
- Better voice quality.

3G

- Internet access.
- Video calling.
- Rise of Mobile applications.

4G (LTE)

- High speed internet.
- Online streaming.
- App-based services (YouTube, WhatsApp).

5G (Current)

- Very high speed.
- Low latency.

- IoT, Smart devices.

Mobile Operating System (Mobile OS)

Mobile Operating System (Mobile OS) is system software that runs on mobile devices like smartphones and tablets. It controls the hardware, manages resources, and provides a platform to run mobile applications.

Main Functions of a Mobile OS

- **Device Management** – controls CPU, memory, battery, camera, sensors.
- **Application Management** – installs, runs, and removes apps.
- **User Interface (UI)** – touch screen, gestures, notifications.
- **Security** – app permissions, data protection.
- **Connectivity** – Wi-Fi, Bluetooth, mobile data, GPS.

Popular Mobile Operating Systems

Android

- Developed by Google
- Open-source
- Used by Samsung, Xiaomi, Vivo, Oppo, etc.

iOS

- Developed by Apple
- Closed-source
- Used only in iPhone & iPad

Why Mobile Applications are Important

- Easy access.
- Portability.
- Real-time services.
- Business growth.
- User engagement.

Overview of Android

Android is an open-source, Linux-based mobile operating system mainly used in smartphones and tablets. It was developed by Google and provides a platform for running mobile applications, managing device hardware, and offering a touch-based user interface.

Features of Android

- **Open Source** – manufacturers and developers can customize it.
- **App Support** – millions of apps available on Google Play Store.
- **User Interface** – touch, gestures, widgets.
- **Security** – app permissions, sandboxing.
- **Connectivity** – Wi-Fi, Bluetooth, GPS, mobile data.

History of Android

- Android Inc. founded in 2003.
- Acquired by Google in 2005.
- First Android version released in 2008.

Open Source Platform

An Open Source Platform is a software platform whose source code is freely available to the public. Anyone can view, use, modify, and distribute the code according to the open-source license.

Characteristics

- Source code is public.
- Modifiable & customizable.
- Community-driven development.
- Usually free or low cost.
- Transparent & secure.
- Android is a popular open-source mobile platform.

Advantages

- No vendor lock-in.
- High flexibility & scalability.
- Large developer community.
- Faster innovation.

Disadvantages

- Requires technical knowledge.
- Limited official support (sometimes).

Open Handset Alliance (OHA)

The Open Handset Alliance (OHA) is a consortium (group) of companies formed to develop open standards for mobile devices. It was established in 2007 and is best known for developing and supporting the Android platform.

Founded By

- Google (leader of OHA)

Members Include

- Mobile device manufacturers
- Software developers
- Semiconductor companies
- Network operators

Main Objectives of OHA

- Develop open mobile platforms.
- Accelerate innovation in mobile technology.
- Reduce development cost.
- Provide better user experience.
- Promote Android as an open-source OS.

Role in Android

- OHA collaboratively developed Android.
- Ensures Android is:

1. Open-source
2. Flexible
3. Device-independent

Why Use Android?

Android is widely used because it is open-source, flexible, user-friendly, and supported by a huge app ecosystem. It suits students, developers, manufacturers, and general users.

Reasons to Use Android

Open Source Platform

4. Android source code is freely available.
5. Manufacturers and developers can customize it easily.

Large App Ecosystem

6. Millions of apps available on Google Play Store.
7. Apps for education, banking, health, entertainment.

High Customization

8. Change themes, widgets, launchers.
9. Control home screen and settings.

Wide Device Support

10. Available on low-cost to premium devices.
11. Used by many brands worldwide.

Developer Friendly

12. Easy app development using Java / Kotlin.
13. Strong community and documentation.

Multitasking Support

14. Run multiple apps simultaneously.
15. Split screen and background services.

Hardware Integration

16. Supports camera, GPS, sensors, Bluetooth, NFC.

Android Architecture

Android Architecture is a layered software stack that shows how Android works internally—from hardware control to user applications.

Layers of Android Architecture (Bottom → Top)

Linux Kernel

17. Core of Android OS.
18. Manages hardware, memory, processes, power, security.
19. Examples: device drivers, camera, display, audio.

Hardware Abstraction Layer (HAL)

20. Interface between hardware and Android framework.
21. Allows Android to work on different hardware without changes.

Native Libraries

22. Written in C/C++.
23. Provide core functionalities.
24. Examples:
    - SQLite (database)
    - WebKit (browser engine)
    - OpenGL (graphics)
    - Media Libraries (audio/video)

Android Runtime (ART)

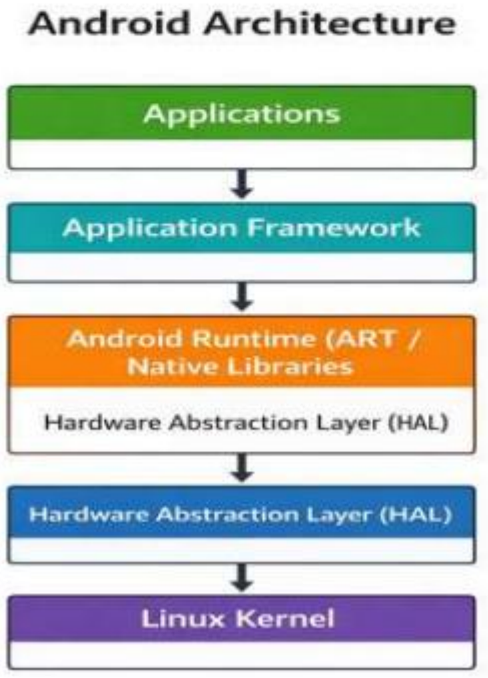
25. Executes Android applications.
26. Includes:
    - Core libraries.
    - Runtime environment.
27. Improves performance and battery efficiency.

Application Framework

28. APIs used by developers.
29. Key components:
    - Activity Manager
    - Window Manager
    - Content Providers
    - Notification Manager
    - Resource Manager

Applications

30. Top layer.
31. Includes:
    - System apps (Phone, SMS, Settings)
    - User-installed apps (WhatsApp, Chrome, etc.)



Android Installation Link

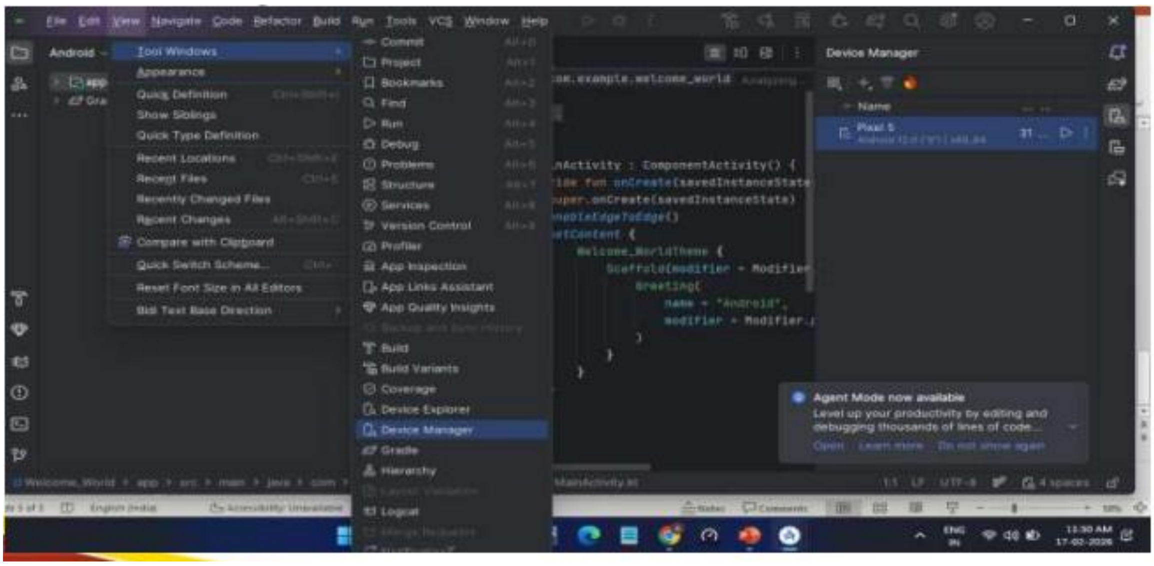
- *https://developer.android.com/studio?utm_source=chatgpt.com*
- This page will automatically let you download the correct installer for your operating system (Windows/macOS/Linux).
- Open the link above.
- Click Download Android Studio.
- Accept the License Agreement.
- Choose the installer for your OS (e.g., Windows).
- Follow the on-screen setup steps.

STEP 1: Emulator (AVD) Setup – Step by Step

- Open Android Studio.
- Launch Android Studio.
- Open Any Project (Hello World or Empty Activity).

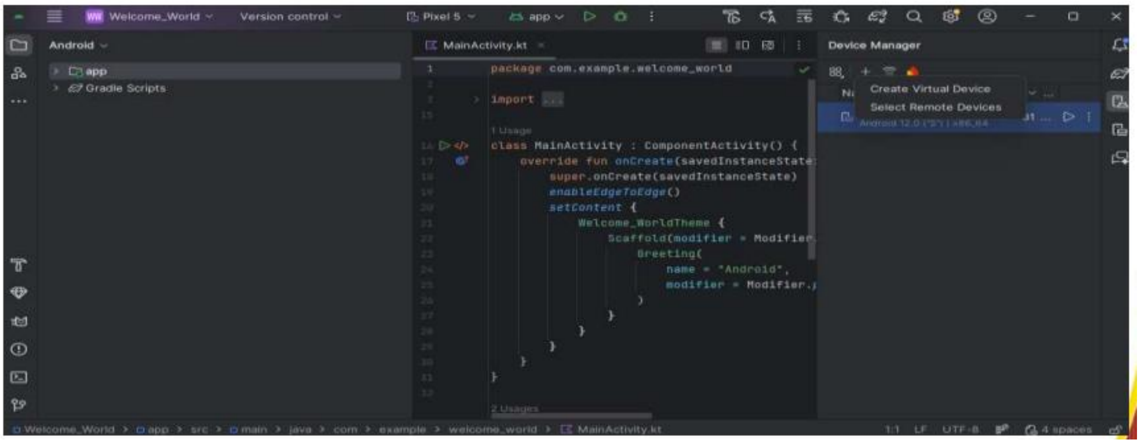
STEP 2: Open Device Manager

- **Tools → Device Manager**
- Click on Device Manager icon.
- Emulator manage from here.



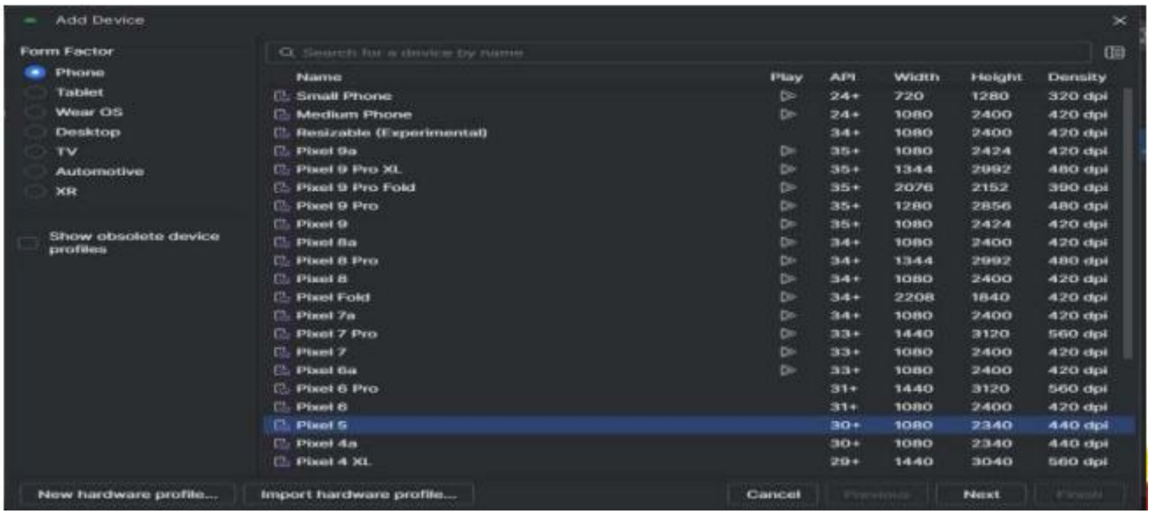
STEP 3: Create Virtual Device

- Click on “Create Device” button.



STEP 4: Select Hardware Device

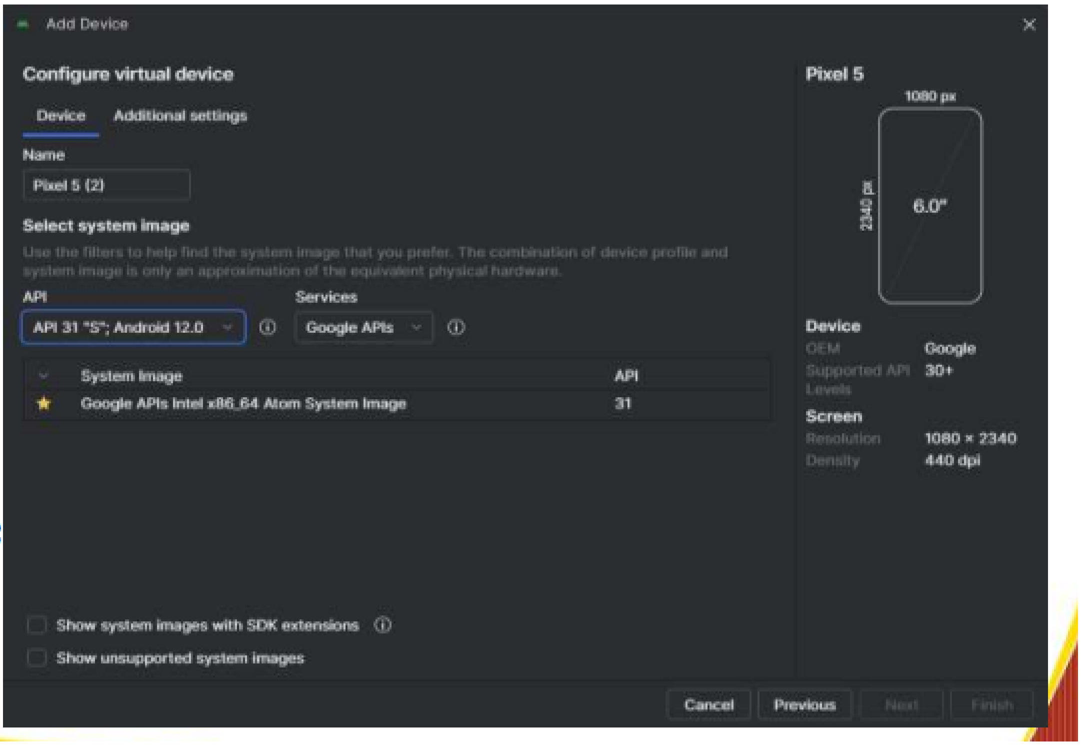
- Select any one.
- Pixel 5 (recommended).
- Pixel 4 / Pixel 6.
- Click Next.



STEP 5: Select System Image

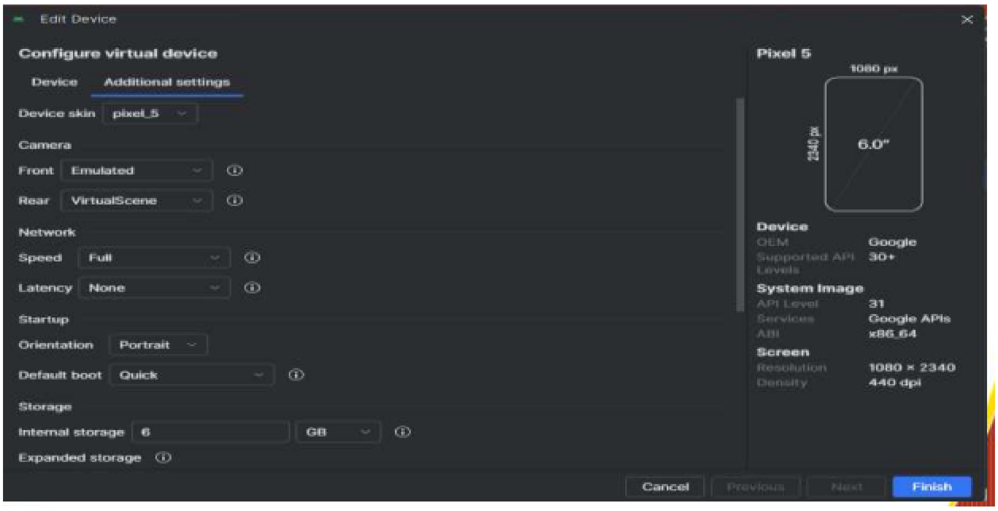
- Choose Android version here.
- **System Image** – means Android operating system.
- Recommended:
    - Android 11 (R)
    - Android 12 (S)
- Click Download.

- Accept License.
- Download complete.
- Click Next.



STEP 6: AVD Configuration

- **AVD Name** – default
- **Orientation** – Portrait.
- **Graphics** – Automatic.
- Click Finish.



STEP 7: Run Emulator

- Click on Device Manager ▶ Play button.
- It will show mobile screen.
- **Emulator Successfully Running = Setup Complete.**

Hello World App

- STEP 1: Click on Android Studio → New Project.
- STEP 2: Select Project Template.

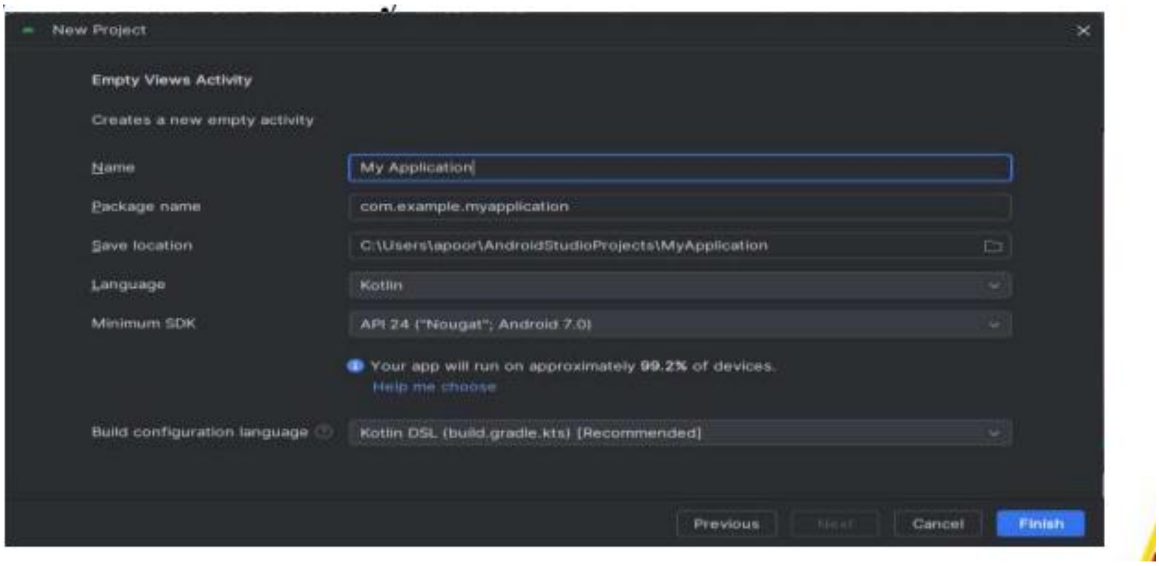
- Select Empty Views Activity.
- Click Next.

STEP 3: Project Configuration

1. Name: HelloWorld
2. Package name: default
3. Language: Java ✅
4. Minimum SDK: API 21 or 23 (default).
5. Load project.

STEP 5: App Run

- Click on green ▶ Run button on Top.
- It will show HelloWorld App on screen.



DEVELOPMENT OF HELLO WORLD ANDROID APPLICATION

Experiment No.: 2

Aim: To develop a simple Hello World Android application using Android Studio.

Software / Hardware Requirement:

- Operating System: Windows 10 / 11
- Android Studio
- JDK (Inbuilt with Android Studio)
- Minimum 8 GB RAM

Theory:

A Hello World application is the first basic program created by beginners to understand the structure and working of an Android application. It displays a simple message on the screen and helps in understanding project creation, layout design, and application execution in Android Studio.

Procedure / Steps:

Step 1: Open Android Studio. Step 2: Click on New

Project.

Step 3: Select Empty Views Activity and click Next. Step 4: Enter

Application Name as Hello World.

Step 5: Select Language as Java.

Step 6: Select Minimum SDK and click Finish.

Step 7: Android Studio will create the project automatically. Step 8: Open activity_main.xml

file.

Step 9: Add TextView and set text as 'Hello World'.

Step 10: Run the application using Emulator or Physical Device.

Program Code:

```
<TextView                 android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Hello     World"     android:textSize="24sp"
   android:layout_centerInParent="true"/>
```

Output:

The application displays the message 'Hello World' on the Android device screen.

Result:

The Hello World Android application was successfully developed and executed.

Conclusion:

This experiment helps in understanding the basics of Android application development, project creation, layout design, and execution using Android Studio.

Android Application: Hello Name App

Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

Step 1: Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:orientation="vertical"
android:padding="16dp">

    <EditText
        android:id="@+id/editName"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter your name" />

    <Button
        android:id="@+id/btnOK"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="OK" />

    <TextView
        android:id="@+id/txtResult"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="18sp"
        android:layout_marginTop="16dp" />

</LinearLayout>
```

Step 2: Java Code (MainActivity.java)

```
package com.example.helloname;
import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText; import
android.widget.TextView;
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText editName;
    Button btnOK;
    TextView txtResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        editName = findViewById(R.id.editName);
btnOK = findViewById(R.id.btnOK);         txtResult
= findViewById(R.id.txtResult);
         btnOK.setOnClickListener(new View.OnClickListener()
{
```

```
            @Override
            public void onClick(View v) {
```



```
                String name = editName.getText().toString();
txtResult.setText("Hello " + name);
            }
        });
    }
}
```

Step 3: Working

- User enters name in EditText
- Clicks OK button
- Program reads input and displays "Hello " in TextView

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DEVELOPMENT OF ANDROID REGISTRATION FORM APPLICATION

Experiment No.: 3

Aim: Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picker), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button.

Theory:

This document provides an example Android application that creates a form with fields including User Name, Password, Address, Gender selection, Age, Date of Birth selection via a DatePickerDialog, and a State selection via a Spinner. On clicking submit, the entered data is retrieved and displayed programmatically.

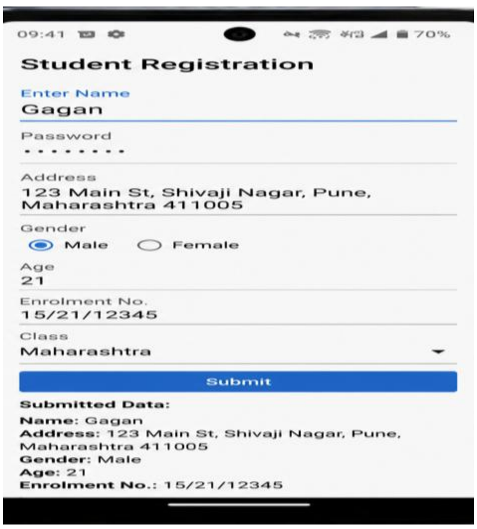
1. XML Layout (activity_main.xml)

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:orientation="vertical"
android:padding="16dp">

    <EditText android:id="@+id/etName" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="User Name"/>
    <EditText android:id="@+id/etPassword" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Password" android:inputType="textPassword"/>
    <EditText android:id="@+id/etAddress" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Address"/>

    <RadioGroup android:id="@+id/rgGender" android:layout_width="match_parent"
android:layout_height="wrap_content" android:orientation="horizontal">
        <RadioButton android:id="@+id/rbMale" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Male" android:checked="true"/>
<RadioButton android:id="@+id/rbFemale" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Female"/>
    </RadioGroup>

    <EditText android:id="@+id/etAge" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Age" android:inputType="number"/>
    <EditText android:id="@+id/etDOB" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Date of Birth" android:focusable="false"/>
    <Spinner android:id="@+id/spState" android:layout_width="match_parent"
android:layout_height="wrap_content"/>

    <Button android:id="@+id/btnSubmit" android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Submit"/>

    <TextView      android:id="@+id/tvResult"      android:layout_width="match_parent"
android:layout_height="wrap_content" android:paddingTop="20dp" android:textSize="16sp"
android:textStyle="bold"/> </LinearLayout>
```

2. MainActivity.java

```
import android.app.DatePickerDialog;
import android.os.Bundle; import
android.view.View; import
android.widget.*;
import androidx.appcompat.app.AppCompatActivity; import
java.util.Calendar;

public class MainActivity extends AppCompatActivity {
EditText name, password, address, age, dob;
```

```
    RadioGroup genderGroup;
    Spinner stateSpinner;
    Button submit;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);         setContentView(R.layout.activity_main);

        name = findViewById(R.id.etName);
password = findViewById(R.id.etPassword);
address = findViewById(R.id.etAddress);
age = findViewById(R.id.etAge);         dob =
findViewById(R.id.etDOB);         genderGroup =
findViewById(R.id.rgGender);         stateSpinner
= findViewById(R.id.spState);         submit =
findViewById(R.id.btnSubmit);         result =
findViewById(R.id.tvResult);

        String[] states = {"Madhya Pradesh", "Maharashtra", "Delhi", "Gujarat"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_spinner_item, states);
        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
stateSpinner.setAdapter(adapter);

        dob.setOnClickListener(v -> {
            Calendar c = Calendar.getInstance();
int y = c.get(Calendar.YEAR);             int m =
c.get(Calendar.MONTH);             int d =
c.get(Calendar.DAY_OF_MONTH);
            DatePickerDialog dp = new DatePickerDialog(MainActivity.this,
                (view, year, month, day) -> dob.setText(day + "/" + (month + 1) + "/" + year),
y, m, d);             dp.show();
        });

        submit.setOnClickListener(v -> {
            int selectedId = genderGroup.getCheckedRadioButtonId();
            RadioButton rb = findViewById(selectedId);
            String gender = rb.getText().toString();
            String data = "Name: " + name.getText().toString() + "\n"
                + "Password: " + password.getText().toString() + "\n"
                + "Address: " + address.getText().toString() + "\n"
                + "Gender: " + gender + "\n"


                    + "Age: " + age.getText().toString() + "\n"
                + "DOB: " + dob.getText().toString() + "\n"
                + "State: " + stateSpinner.getSelectedItem().toString();
result.setText(data);
        });
    }
}
```



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DEVELOPMENT OF ANDROID INTENT APPLICATION

Experiment No.: 4
Aim: Design an android application to create page using Intent and one Button and pass the Values from one Activity to second Activity.
Theory:
An Intent is a messaging object you can use to request an action from another app component. In this experiment, we use an Explicit Intent to start a second activity and pass data using the putExtra() method. The receiving activity retrieves this data using getIntent().getStringExtra().

1. XML Layouts

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:orientation="vertical"
android:gravity="center"     android:padding="20dp">
    <EditText
        android:id="@+id/editTextName"
android:layout_width="match_parent"
android:layout_height="wrap_content"         android:hint="Enter
Name" />
    <Button
        android:id="@+id/buttonSend"
android:layout_width="wrap_content"
android:layout_height="wrap_content"         android:text="Send"
        android:layout_marginTop="20dp" />
</LinearLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:orientation="vertical"
android:gravity="center">
    <TextView
        android:id="@+id/textViewResult"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="24sp" /> </LinearLayout>
```

2. Java Source Code

MainActivity.java

```
package com.example.intentdemo;
import android.content.Intent;
import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText editText;
    Button button;
@Override     protected void
onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstance
State);
setContentView(R.layout.acti
```

```
vity_main);          editText
=
findViewById(R.id.editTextNa
me);          button =
findViewById(R.id.buttonSend
);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = editText.getText().toString();
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                // Passing value using Intent
intent.putExtra("username", name);
startActivity(intent);
            }
        });
}
}
```

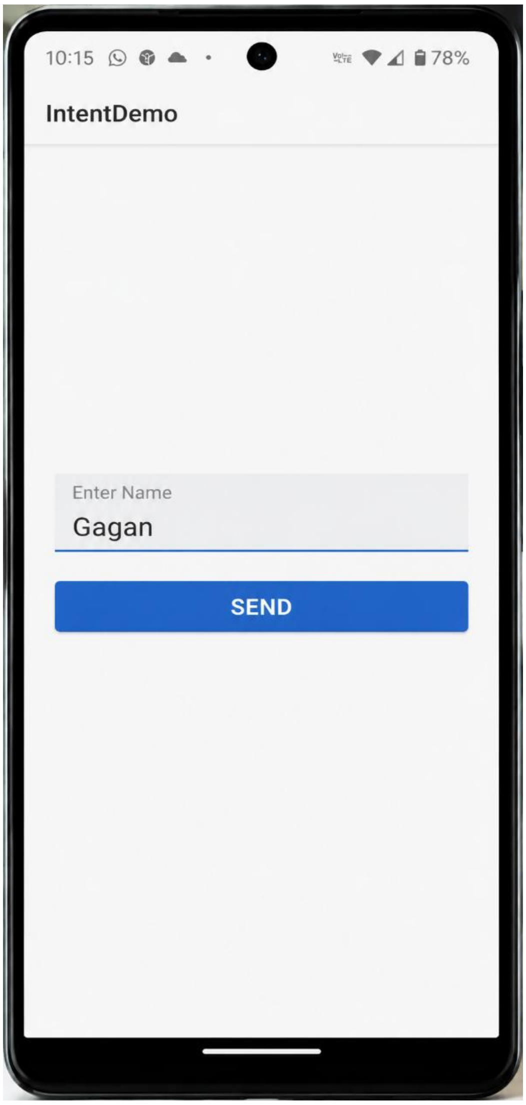
SecondActivity.java

```
package com.example.intentdemo;
import android.os.Bundle; import
android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    TextView textView;
@Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_second);         textView
= findViewById(R.id.textViewResult);
        // Receiving data from Intent
        String data = getIntent().getStringExtra("username");
textView.setText("Welcome " + data);
    }
}
```

3. AndroidManifest.xml

```
<application ... >
    <activity android:name=".SecondActivity"></activity>
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```


10:15
78%
IntentDemo
Enter Name
Gagan
SEND


10:15
78%
IntentDemo
Welcome Gagan

Create an android application using Fragments.

```
// MainActivity.java package
com.example.fragmentdemo;  import
android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    }
}
//   FirstFragment.java   package
com.example.fragmentdemo;        import
android.os.Bundle;               import
android.view.LayoutInflater;     import
android.view.View;               import
android.view.ViewGroup;          import
androidx.annotation.NonNull;     import
androidx.annotation.Nullable;    import
androidx.fragment.app.Fragment;

public class FirstFragment extends Fragment {
public FirstFragment() {
        // Required empty public constructor
    }

    @Nullable
@Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
@Nullable Bundle savedInstanceState) {         return
inflater.inflate(R.layout.fragment_first, container, false);     }
}
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"  android:orientation="vertical">
 <fragment
 android:id="@+id/fragment"
 android:name="com.example.fragmentdemo.FirstFragment"
android:layout_width="match_parent"
android:layout_height="match_parent" />
</LinearLayout>
<!-- fragment_first.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:gravity="center"  android:orientation="vertical">
 <TextView
 android:layout_width="wrap_content"
android:layout_height="wrap_content"
```

```
android:text="This is Fragment Example"  android:textSize="24sp" />
</LinearLayout>
<!-- AndroidManifest.xml -->
<application
 ... >
 <activity android:name=".MainActivity">
 <intent-filter>
 <action android:name="android.intent.action.MAIN" />
 <category android:name="android.intent.category.LAUNCHER" />
 </intent-filter>
 </activity>
</application>
```

Design an android application Send SMS using Intent

XML Code (activity_main.xml)

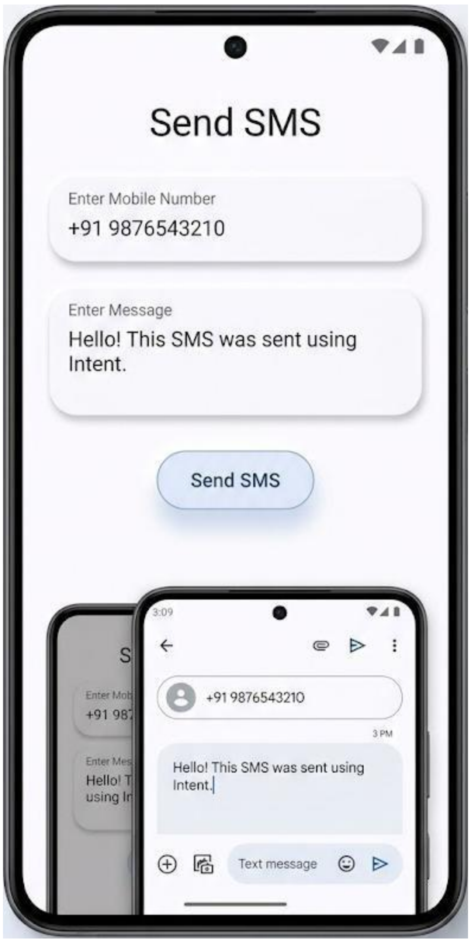
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"  android:padding="20dp">
 <EditText
 android:id="@+id/etNumber"
android:layout_width="match_parent"
android:layout_height="wrap_content"  android:hint="Enter
Mobile Number"  android:inputType="phone"/>
 <EditText
 android:id="@+id/etMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"  android:hint="Enter
Message"  android:inputType="textMultiLine"/>
 <Button
 android:id="@+id/btnSend"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Send SMS"/> </LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.smsintent;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent; import
android.net.Uri; import android.os.Bundle;
import android.view.View; import
android.widget.Button; import
android.widget.EditText;

public class MainActivity extends AppCompatActivity {
 EditText etNumber, etMessage;
 Button btnSend;
@Override
 protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);  etNumber =
findViewById(R.id.etNumber);  etMessage =
findViewById(R.id.etMessage);  btnSend =
findViewById(R.id.btnSend);
 btnSend.setOnClickListener(new View.OnClickListener() {
 @Override
 public void onClick(View v) {
 String number = etNumber.getText().toString();
 String message = etMessage.getText().toString();
 Intent intent = new Intent(Intent.ACTION_VIEW);
intent.setData(Uri.parse("sms:" + number));
intent.putExtra("sms_body", message);
startActivity(intent);  }
```

```
        });
        }
      }
```


Send SMS
Enter Mobile Number
+91 9876543210
Enter Message
Hello! This SMS was sent using Intent.
Send SMS
3:09
+91 9876543210
3 PM
Hello! This SMS was sent using Intent.
Text message

Design an android application Using Radiobuttons management.

XML Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"  android:padding="20dp">
 <TextView
 android:layout_width="wrap_content"
android:layout_height="wrap_content"  android:text="Select
Your Department"  android:textSize="20sp"/>
 <RadioGroup
 android:id="@+id/radioGroup"
android:layout_width="wrap_content"
android:layout_height="wrap_content">
 <RadioButton
android:id="@+id/rbCSE"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="CSE"/>  <RadioButton
android:id="@+id/rbIT"
 android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="IT"/>  <RadioButton
android:id="@+id/rbECE"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="ECE"/>
 </RadioGroup>
<Button
 android:id="@+id/btnSubmit"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Submit"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package
com.example.radiobuttondemo;                    import
androidx.appcompat.app.AppCompatActivity; import
android.os.Bundle;   import   android.view.View;
import       android.widget.Button;       import
android.widget.RadioButton;                 import
android.widget.RadioGroup;                  import
android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    RadioGroup radioGroup;
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
                                                super.onCreate(savedInstanceState);
                                            setContentView(R.layout.activity_main);

        radioGroup      =      findViewById(R.id.radioGroup);
btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int selectedId = radioGroup.getCheckedRadioButtonId();
                RadioButton radioButton = findViewById(selectedId);
                Toast.makeText(MainActivity.this,
                               "Selected: " + radioButton.getText(),
                               Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```